

Chapter 5: Cross Sections and Decay Rates

Quantum Field Theory and the Standard Model

In Jung Kim

5. Introduction: Collider Physics

- **The 20th Century Legacy:**
 - Invention and development of collider physics as an efficient way to determine which particles exist, their properties, and how they interact.
 - Historical milestones:
 - **Rutherford (1911):** Discovered the nucleus using α -particles from nature.
 - **Anderson (1932):** Discovered the positron from cosmic rays.
 - **Lawrence (1931):** Built a 4-inch cyclotron giving protons 80,000 eV, paving the way for relativistic laboratory beams.
 - **Large Hadron Collider (LHC):** Collides proton beams at 7 TeV (and more).

5. Why Colliders?

- **Fundamental Interactions Study:**

- **Initial States:** Essentially fixed momenta, i.e., plane waves.
- **Final States:** Fixed momenta, measured at $t = +\infty$.
- **Goal:** By measuring the mapping from initial momenta to final momenta, we compare experiment with theoretical models like Quantum Field Theory (QFT).

- **Role of S -matrix:**

- S -matrix elements contain all time-evolution information:

$$\langle f|S|i\rangle_{\text{Heisenberg}} = \langle f; \infty|i; -\infty\rangle_{\text{Schrödinger}} \quad (5.1)$$

- Assumes asymptotic states at $t = \pm\infty$ are free of interactions.

5.1 Cross Sections: Classical Definition

- **Rutherford's Motivation:** Determining the size r of an atomic nucleus.
- **Classical Scattering Area** ($\sigma = \pi r^2$):

$$\sigma = \frac{\text{number of particles scattered}}{\text{time} \times \text{number density in beam} \times \text{velocity of beam}} = \frac{1}{T} \frac{1}{\Phi} N \quad (5.2)$$

- T : Time of the experiment.
- Φ : Incoming flux ($\Phi = \text{number density} \times \text{velocity of beam}$).
- N : Number of particles scattered.
- **Factors of Experiment:** Flux (Φ) and Time (T) depend on experimental setup, but the number of scatterings (N) is determined by **short-distance particle interactions**.

5.1 Differential Cross Section & Probability

- **Differential Cross Section ($d\sigma/d\Omega$):**
 - Gives the number of scattered particles in a certain solid angle $d\Omega$.
 - Classically gives the shape of the object or form of the potential.
- **Quantum Mechanical Generalization:**
 - Generalizes the notion of area to a measure of interaction strength.
 - Classical differential probability is $P = \frac{N}{N_{\text{inc}}}$.
 - Quantum mechanical cross section:

$$d\sigma = \frac{1}{T} \frac{1}{\Phi} dP \quad (5.3)$$

- Φ is now normalized as if the beam has just **one particle**.
- P is the quantum mechanical probability of scattering.
- Differential number of events: where $\mathcal{L}$ is luminosity (5.4).

$$dN = \mathcal{L} \times d\sigma$$

5.1 Relating $d\sigma$ to S -matrix Elements

- We focus on the $2 \rightarrow n$ process:

$$p_1 + p_2 \rightarrow \{p_j\} \quad (5.5)$$

- **Incoming Flux (Φ) in Different Frames:**

- In target rest frame: $\Phi = \frac{|\vec{v}|}{V}$.
- In general frame (e.g., Center-of-Mass frame):

$$\Phi = \frac{|\vec{v}_1 - \vec{v}_2|}{V}$$

- **Differential Cross Section:**

$$d\sigma = \frac{V}{T} \frac{1}{|\vec{v}_1 - \vec{v}_2|} dP \quad (5.6)$$

- **Normalized Differential Probability:**

$$dP = \frac{|\langle f|S|i\rangle|^2}{\langle f|f\rangle\langle i|i\rangle} d\Pi \quad (5.7)$$

- $d\Pi$: Region of final-state momenta, $d\Pi = \prod_j \frac{V}{(2\pi)^3} d^3p_j \quad (5.8)$.

5.1 State Normalization in Finite Volume

- In Chapter 2, state normalization was defined as:

$$a_k^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_k}} |k\rangle \quad (5.9)$$

$$[a_p, a_q^\dagger] = (2\pi)^3 \delta^3(p - q)$$

- This leads to the inner product:

$$\langle p|p\rangle = (2\pi)^3 (2\omega_p) \delta^3(0) \quad (5.10)$$

- Regulating the infinite $\delta^3(0)$ using finite volume V :

$$(2\pi)^3 \delta^3(p) = \int d^3x e^{i\vec{p}\cdot\vec{x}} \implies \delta^3(0) = \frac{V}{(2\pi)^3} \quad (5.11, 5.12)$$

- Similarly, for time T :

$$\delta^4(0) = \frac{TV}{(2\pi)^4} \quad (5.13)$$

5.1 Inner Products & S -matrix Structure

- **Single-particle state normalization:**

$$\langle p|p\rangle = 2\omega_p V = 2E_p V \quad (5.14)$$

- **Initial and final state normalization:**

$$|i\rangle = |p_1\rangle|p_2\rangle \implies \langle i|i\rangle = (2E_1 V)(2E_2 V) \quad (5.15)$$

$$|f\rangle = \prod_j |p_j\rangle \implies \langle f|f\rangle = \prod_j (2E_j V) \quad (5.15)$$

- **The Transfer Matrix (T):**

- In a free theory, $S = 1$. To describe deviations, we write:

$$S = 1 + iT \quad (5.16)$$

- Factoring out the overall momentum-conserving delta function:

$$T = (2\pi)^4 \delta^4(\Sigma p) \mathcal{M} \quad (5.17)$$

$$\langle f|S - 1|i\rangle = i(2\pi)^4 \delta^4(\Sigma p) \langle f|\mathcal{M}|i\rangle \quad (5.18)$$

5.1 Squaring the Delta Function

- For $f \neq i$, taking the square of the S -matrix element:

$$|\langle f|S|i\rangle|^2 = [(2\pi)^4\delta^4(\Sigma p)]^2 |\langle f|\mathcal{M}|i\rangle|^2$$

- One delta function enforces momentum conservation, while the other gives:

$$(2\pi)^4\delta^4(0) = TV$$

- Thus:

$$|\langle f|S|i\rangle|^2 = \delta^4(\Sigma p)TV(2\pi)^4|\mathcal{M}|^2 \quad (5.19)$$

- Here, $|\mathcal{M}|^2 \equiv |\langle f|\mathcal{M}|i\rangle|^2$.

5.1 Differential Probability dP

- Substituting the normalizations and squared amplitude into dP :

$$dP = \frac{\delta^4(\Sigma p) T V (2\pi)^4}{(2E_1 V)(2E_2 V)} \prod_j \frac{1}{2E_j V} |\mathcal{M}|^2 \prod_j \frac{V}{(2\pi)^3} d^3 p_j$$

- Simplifying all factors of V :

$$dP = \frac{T}{V} \frac{1}{(2E_1)(2E_2)} |\mathcal{M}|^2 d\Pi_{\text{LIPS}} \quad (5.20)$$

- **Lorentz-Invariant Phase Space (LIPS):**

$$d\Pi_{\text{LIPS}} \equiv \left(\prod_{\text{final } j} \frac{d^3 p_j}{(2\pi)^3} \frac{1}{2E_j} \right) (2\pi)^4 \delta^4(\Sigma p) \quad (5.21)$$

- $d\Pi_{\text{LIPS}}$ is completely Lorentz invariant.

5.1 Final Cross Section Formula

- Combining dP with Eq. (5.6), all V and T factors drop out:

$$d\sigma = \frac{1}{2E_1 2E_2 |\vec{v}_1 - \vec{v}_2|} |\mathcal{M}|^2 d\Pi_{\text{LIPS}} \quad (5.22)$$

- Note that the velocity is given by $\vec{v} = \vec{p}/p_0 = \vec{p}/E$.
- This is the fundamental formula relating Feynman-diagram-calculated invariant amplitude $\mathcal{M}$ to physical cross sections!

5.1.1 Decay Rates

- **Definition:** The probability that a one-particle state with momentum p_1 decays into a multi-particle state with momenta $\{p_j\}$ over a time T :

$$d\Gamma = \frac{1}{T} dP \quad (5.23)$$

- Calculated in perturbation theory assuming interactions happen over finite time T , making it equivalent to a $1 \rightarrow n$ scattering process.
- **Decay Rate Formula:**

$$d\Gamma = \frac{1}{2E_1} |\mathcal{M}|^2 d\Pi_{\text{LIPS}} \quad (5.24)$$

- This rate is in the **rest frame** of the decaying particle. For relativistic velocities, time dilation slows down the decay rate ($\Gamma_{\text{boosted}} = \frac{m}{E} \Gamma_{\text{rest}}$).

5.1.2 Special Cases: $2 \rightarrow 2$ CM Frame

- Consider $2 \rightarrow 2$ scattering in the center-of-mass (CM) frame:

$$p_1 + p_2 \rightarrow p_3 + p_4 \quad (5.25)$$

- $\vec{p}_1 = -\vec{p}_2$ and $\vec{p}_3 = -\vec{p}_4$
- $E_1 + E_2 = E_3 + E_4 = E_{\text{CM}}$

- LIPS element:

$$d\Pi_{\text{LIPS}} = (2\pi)^4 \delta^4(\Sigma p) \frac{d^3\vec{p}_3}{(2\pi)^3 2E_3} \frac{d^3\vec{p}_4}{(2\pi)^3 2E_4} \quad (5.26)$$

- Integrate over $d^3\vec{p}_4$ using 3D delta function:

$$d\Pi_{\text{LIPS}} = \frac{1}{16\pi^2} d\Omega \int_0^\infty dp_f \frac{p_f^2}{E_3 E_4} \delta(E_3 + E_4 - E_{\text{CM}}) \quad (5.27)$$

- $p_f = |\vec{p}_3| = |\vec{p}_4|$, $E_3 = \sqrt{m_3^2 + p_f^2}$, $E_4 = \sqrt{m_4^2 + p_f^2}$.

5.1.2 Variable Change & Jacobian

- Define $x(p_f) \equiv E_3(p_f) + E_4(p_f) - E_{\text{CM}}$.
- The Jacobian for variable change:

$$\frac{dx}{dp_f} = \frac{d}{dp_f}(E_3 + E_4 - E_{\text{CM}}) = \frac{p_f}{E_3} + \frac{p_f}{E_4} = \frac{E_3 + E_4}{E_3 E_4} p_f \quad (5.28)$$

- Because of the delta function, $E_3 + E_4 = E_{\text{CM}}$:

$$\frac{dx}{dp_f} = \frac{E_{\text{CM}} p_f}{E_3 E_4} \implies dp_f = dx \frac{E_3 E_4}{p_f E_{\text{CM}}}$$

- Substituting back:

$$\begin{aligned} d\Pi_{\text{LIPS}} &= \frac{1}{16\pi^2} d\Omega \int dx \left(\frac{E_3 E_4}{p_f E_{\text{CM}}} \right) \frac{p_f^2}{E_3 E_4} \delta(x) \\ d\Pi_{\text{LIPS}} &= \frac{1}{16\pi^2} d\Omega \frac{p_f}{E_{\text{CM}}} \theta(E_{\text{CM}} - m_3 - m_4) \quad (5.29) \end{aligned}$$

5.1.2 Differential Cross Section in CM Frame

- Substituting $d\Pi_{\text{LIPS}}$ into $d\sigma$ (Eq. (5.22)):

$$d\sigma = \frac{1}{2E_1 2E_2 |\vec{v}_1 - \vec{v}_2|} \frac{1}{16\pi^2} d\Omega \frac{p_f}{E_{\text{CM}}} |\mathcal{M}|^2 \theta(E_{\text{CM}} - m_3 - m_4) \quad (5.30)$$

- Relative velocity in the CM frame:

$$|\vec{v}_1 - \vec{v}_2| = \left| \frac{|\vec{p}_1|}{E_1} + \frac{|\vec{p}_2|}{E_2} \right| = \frac{|\vec{p}_i| E_{\text{CM}}}{E_1 E_2} \quad (5.31)$$

◦ $p_i = |\vec{p}_1| = |\vec{p}_2|$.

- Substituting flux:

$$2E_1 2E_2 |\vec{v}_1 - \vec{v}_2| = 4|\vec{p}_i| E_{\text{CM}}$$

5.1.2 Final CM Cross Section Formulas

- **General $2 \rightarrow 2$ CM Cross Section:**

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} = \frac{1}{64\pi^2 E_{\text{CM}}^2} \frac{|\vec{p}_f|}{|\vec{p}_i|} |\mathcal{M}|^2 \theta(E_{\text{CM}} - m_3 - m_4) \quad (5.32)$$

- **Equal Mass Limit:**

- If all masses are equal ($m_1 = m_2 = m_3 = m_4$), then $|\vec{p}_f| = |\vec{p}_i|$:

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} = \frac{1}{64\pi^2 E_{\text{CM}}^2} |\mathcal{M}|^2 \quad (5.33)$$

5.2 Non-Relativistic Limit: Born Approximation

- In non-relativistic quantum mechanics, the cross section is given by the **Born approximation**:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Born}} = \frac{m_e^2}{4\pi^2} |\tilde{V}(\vec{k})|^2 \quad (5.34)$$

- $\tilde{V}(\vec{k})$ is the Fourier transform of the potential:

$$\tilde{V}(\vec{k}) = \int d^3x e^{-i\vec{k}\cdot\vec{x}} V(\vec{x}) \quad (5.35)$$

- $\vec{k} = \vec{p}_i - \vec{p}_f$ is the **momentum transfer**.

- **Coulomb Potential Example:**

$$V(\vec{x}) = \frac{e^2}{4\pi|\vec{x}|} \implies \tilde{V}(\vec{k}) = \frac{e^2}{\vec{k}^2}$$
$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Born}} = \frac{m_e^2}{4\pi^2} \left(\frac{e^2}{\vec{k}^2}\right)^2 = \frac{e^4 m_e^2}{4\pi^2 \vec{k}^4} \quad (5.36)$$

5.2 Non-Relativistic Limit: Field Theory Version

- For an electron (m_e) scattering off a heavy proton (m_p), the CM frame is approximately the proton rest frame:
 - $E_{\text{CM}} \approx m_p$.
 - Elastic scattering: $|\vec{p}_i| = |\vec{p}_f|$.
- Field theory CM cross section prediction:

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} = \frac{1}{64\pi^2 m_p^2} |\mathcal{M}|^2 \quad (5.37)$$

- **Dimension of $\mathcal{M}$:**
 - Since $[d\sigma/d\Omega] = -2$ and $[m_p^{-2}] = -2$, $\mathcal{M}$ must be **dimensionless**.

5.2 Scalar QED Lagrangian

- Ignoring spin, the interaction between electron (ϕ_e), proton (ϕ_p), and photon (A_μ) is described by the **Scalar QED Lagrangian**:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 - \phi_e^*(\square + m_e^2)\phi_e - \phi_p^*(\square + m_p^2)\phi_p - ieA^\mu(\phi_e^*\partial_\mu\phi_e - (\partial_\mu\phi_e^*)\phi_e) + ieA^\mu(\phi_p^*\partial_\mu\phi_p - (\partial_\mu\phi_p^*)\phi_p) + \mathcal{O}(e^2) \quad (5.38)$$

- **Non-Relativistic Limit:**

- Momenta are close to rest: $p^\mu = (E, \vec{p}) \sim (m, \vec{0})$.
- $E \sim m \implies \partial_t\phi \sim im\phi$, and $|\vec{p}| \ll m$.
- Leading-order time dependence factorized out:

$$\phi_e \rightarrow \phi_e e^{-im_e t}, \quad \phi_p \rightarrow \phi_p e^{-im_p t}$$

5.2 Non-Relativistic Lagrangian & Guessing $\mathcal{M}$

- Substituting the time dependence factorization into Eq. (5.38):

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + \phi_e^* \vec{\nabla}^2 \phi_e + 2em_e A_0 \phi_e^* \phi_e + \phi_p^* \vec{\nabla}^2 \phi_p - 2em_p A_0 \phi_p^* \phi_p + \dots \quad (5.39)$$

- Here, dots represent higher orders in $\frac{\vec{\nabla}^2}{m^2}$.
- **Guessing the Invariant Amplitude $\mathcal{M}$:**
 - Proton-photon interaction: $\propto -2em_p$
 - Photon propagator: $\propto \frac{1}{\vec{k}^2}$
 - Electron-photon interaction: $\propto 2em_e$
 - Therefore:

$$\mathcal{M} \sim (-2em_p) \frac{1}{\vec{k}^2} (2em_e) = -\frac{4e^2 m_e m_p}{\vec{k}^2} \quad (5.40)$$

5.2 Normalization Rescaling

- Substituting $\mathcal{M}$ into the cross section formula (Eq. (5.37)):

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} \approx \frac{1}{64\pi^2 m_p^2} \left(\frac{16e^4 m_e^2 m_p^2}{\vec{k}^4} \right) = \frac{e^4 m_e^2}{4\pi^2 \vec{k}^4} \quad (5.41)$$

- This matches Born approximation (Eq. (5.36))!
- **Relating Relativistic & Non-Relativistic Conventions:**
 - The factor m in interaction terms of Eq. (5.39) is unconventional.
 - Rescaling fields standardizes the kinetic terms:

$$\phi \rightarrow \frac{1}{\sqrt{2m}} \phi \quad \Longrightarrow \quad \mathcal{L}_{\text{rescaled}} = -\frac{1}{4} F_{\mu\nu}^2 + \frac{1}{2m_e} \phi_e^* \vec{\nabla}^2 \phi_e + \frac{1}{2m_p} \phi_p^* \vec{\nabla}^2 \phi_p + eA_0 \phi_e^* \phi_e - eA_0 \phi_p^* \phi_p \quad (5.42)$$

- Note that since $\frac{1}{\sqrt{2m}} \approx \frac{1}{\sqrt{2E}}$ in the NR limit, this is directly connected to the $\frac{1}{\sqrt{2\omega_p}}$ factors in the definition of quantum fields.

5.3 $e^+e^- \rightarrow \mu^+\mu^-$ with Spin

- **Spin 1/2 Considerations:**

- Coulomb's law ($1/r$) does not involve spin at leading order (due to Gauss's law/dimensional analysis).
- However, relativistic spin can be computed via spin projections.

- **Process:** e^+e^- annihilation into virtual photon γ^* , which decays to $\mu^+\mu^-$.

$$A_{e^-e^+\rightarrow\gamma\rightarrow\mu^-\mu^+} \quad (5.43)$$

- **Dimensional Analysis:**

- Propagator: $\frac{1}{k^2} = \frac{1}{E_{\text{CM}}^2}$.
- The $1/E_{\text{CM}}^2$ is canceled by state normalization factors:

$$\sqrt{2E_1}\sqrt{2E_2}\sqrt{2E_3}\sqrt{2E_4} = E_{\text{CM}}^2$$

- This leaves $\mathcal{M}$ as a dimensionless number from spin projections.

5.3 Spin Projections and Polarizations

- **Invariant Amplitude with Spin Projections:**

$$\mathcal{M}(s_1 s_2 \rightarrow s_3 s_4) = \sum_{\epsilon} \langle s_1 s_2 | \epsilon \rangle \langle \epsilon | s_3 s_4 \rangle \quad (5.44)$$

- **Kinematics at Ultra-Relativistic Energies ($E \gg m_e, m_\mu$):**

- Neglect particle masses.
- Along z -axis, initial momenta are:

$$p_1^\mu = (E, 0, 0, E), \quad p_2^\mu = (E, 0, 0, -E) \quad (5.45)$$

- **Initial Spin States:**

- Better expressed using polarization/helicity rather than up/down.
- 4 possible initial spin states:

$$|s_1 s_2\rangle = | \leftrightarrow \leftrightarrow \rangle, \quad | \uparrow \uparrow \rangle, \quad | \uparrow \leftrightarrow \rangle, \quad | \leftrightarrow \uparrow \rangle \quad (5.46)$$

5.3 Photon & Final State Muon Polarizations

- Photon has spin 1 and two physical polarizations:
 - Only $|\leftrightarrow\leftrightarrow\rangle$ and $|\uparrow\downarrow\rangle$ can annihilate into a photon to conserve angular momentum.
 - Photon polarizations produced:

$$\epsilon_1^\mu = (0, 1, 0, 0), \quad \epsilon_2^\mu = (0, 0, 1, 0) \quad (5.47)$$

- **Final Muon/Antimuon States:**

- Momenta at angle θ relative to z -axis:

$$p_3^\mu = E(1, 0, \sin \theta, \cos \theta), \quad p_4^\mu = E(1, 0, -\sin \theta, -\cos \theta) \quad (5.48)$$

- Corresponding photon polarizations (orthogonal to p_3 and p_4):

$$\bar{\epsilon}_1^\mu = (0, 1, 0, 0), \quad \bar{\epsilon}_2^\mu = (0, 0, \cos \theta, -\sin \theta) \quad (5.49)$$

5.3 Matrix Elements & Final Cross Section

- **Non-vanishing Spin Amplitudes:**

$$\mathcal{M}_1 = \mathcal{M}(|\leftrightarrow\leftrightarrow\rangle \rightarrow |f\rangle) = \epsilon_1 \cdot \bar{\epsilon}_1 + \epsilon_1 \cdot \bar{\epsilon}_2 = -1 \quad (5.50)$$

$$\mathcal{M}_2 = \mathcal{M}(|\uparrow\uparrow\rangle \rightarrow |f\rangle) = \epsilon_2 \cdot \bar{\epsilon}_1 + \epsilon_2 \cdot \bar{\epsilon}_2 = -\cos\theta \quad (5.51)$$

- **Unpolarized Cross Section (Average & Sum):**

- Averaging over initial states and summing over final states:

$$|\mathcal{M}|^2 = |\mathcal{M}_1|^2 + |\mathcal{M}_2|^2 = 1 + \cos^2\theta \quad (5.52)$$

- **Final Differential Cross Section:**

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{64\pi^2 E_{\text{CM}}^2} (1 + \cos^2\theta) \quad (5.53)$$

- This is the exact tree-level QED result, fully matching Chapter 13!